RICARDO CHARLES

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PROFESSIONAL EXPERIENCE

AI Software Engineer | AI/ML SME **02/2025 - Present**

Systems Planning and Analysis, Inc., Alexandria, VA

- Designed and built an autonomous AI agent system for intelligent document processing using C#/.NET 8 and Microsoft Semantic Kernel, orchestrating multi-step workflows with Azure OpenAI and supporting local Ollama/Gemma inference for restricted environments.
- Developed an MCP-based document preprocessing service that parsed and transformed source files into normalized JSON representations, creating a standardized handoff to an AI ingestion pipeline and reducing downstream document processing overhead.
- Developed an end-to-end Retrieval-Augmented Generation (RAG) pipeline for document parsing, sentence-aware chunking, batch embedding, vector storage, retrieval, interactive Q&A, and large-document summarization.
- Designed a modular vector search layer using SQLite and SIMD-accelerated cosine similarity, enabling migration to Qdrant or pgvector without changes to the broader RAG pipeline.
- Led an evaluation of nine agentic AI coding harnesses and hosted/local LLM backends against infrastructure constraints, identified failure modes and token-cost impacts, and authored a white paper whose recommendation was adopted by the team.
- Provide subject matter expertise in data science and AI/ML to government clients, advising on technology adoption, system modernization, knowledge graphs, cloud migration, and mission-focused solutions.
- Advise program managers and senior leadership on AI/ML strategy, technical feasibility, DoD risk management requirements, ML lifecycle management, responsible AI, and alignment with mission objectives.

Data Scientist **03/2024 - 11/2024**

Booz Allen Hamilton, McLean, VA

- Enhanced data analysis scalability by converting SAS templates into Jupyter Notebooks, leveraging PostgreSQL and Elasticsearch to optimize data storage, retrieval, and query performance. This transformation enabled efficient, scalable data access for real-time analytics, streamlining workflows and improving cross-functional data integration.
- Drove data-driven decision-making by employing Python for data manipulation, predictive modeling, and visualization. Applied machine learning algorithms, including regression, classification, and clustering, to extract actionable insights, utilizing libraries like NumPy, pandas, and scikit-learn to ensure high accuracy and reliability in data outcomes.

Cyber Threat Engineer | Data Scientist **06/2021 - 02/2024**

MITRE, McLean, VA

- **Built Web Scraping Data Pipelines and Counterfeit Detection Models:** Developed custom Python web crawlers that collected product listing data from online marketplaces and official retail sites; cleaned, structured, and analyzed the data and flagged counterfeit and fraudulent goods using rule-based scoring and a classification model. Delivered findings through dashboards and reports, and continuously updated the crawlers as sites changed to maintain data quality.
- **Developed Blockchain Transaction Analysis Tooling for Asset Recovery:** Designed and implemented tooling that analyzed blockchain transaction data to trace wallets and follow the flow of funds, enabling the seizure of cryptocurrency assets from criminal actors. Performed rigorous regression testing to ensure reliable performance in live environments.
- **Analyzed IoT Network Traffic Data for Anomalies:** Configured and maintained a fleet of IoT devices using Python and JavaScript; captured device network traffic with Wireshark and built Python scripts and visualizations to detect patterns and anomalies, identifying and blocking unauthorized outbound communications to external servers.

Software Developer **01/2019 - 01/2020**

Freddie Mac, McLean, VA

- **Developed High-Quality, Reliable Web Application Code:** Crafted and maintained robust code for a web interface that integrates data, analytics, and reporting, ensuring smooth functionality. Implemented automated testing (Unit, Functional, Performance, and Acceptance) to guarantee stability and eliminate bugs, resulting in a seamless, error-free application experience for end users.

- **Collaborated on Solution Design and Documentation:** Partnered with senior team members to document project requirements, design solutions, and implement them effectively. Helped build clear, maintainable workflows and interactive tools that track user activity and provide a comprehensive audit trail, enhancing system transparency.

SKILLS

Languages: Python, Java, C++, C#, Bash

Machine Learning: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy

Generative AI: Azure OpenAI, Microsoft Semantic Kernel, Ollama, Gemma, RAG, Agentic AI, Embeddings, Vector Search, Prompt Engineering, Tool Calling

Software/Tools: MCP, LangGraph, OpenAI Agents SDK, Ragas, vLLM, FastAPI, pytest, Qdrant, pgvector, Jupyter Notebooks, PostgreSQL, MySQL, Elasticsearch, MongoDB, SQL, MongoDB Compass, Git, Docker, Angular, Spring Boot, Postman, Jenkins, Gradle/Maven, Cucumber, Wireshark

Methodology: Agile, Modern Delivery, Test-Driven Development (TDD)

Operating Systems: Linux, Windows, macOS

PROJECTS

LLM Evaluation & Observability Platform
Python, FastAPI, Ragas, Langfuse

- Built an automated evaluation platform for LLM, RAG, and agentic AI applications, measuring response quality, retrieval accuracy, latency, token usage, and cost.
- Implemented end-to-end observability and tracing for model calls, retrieval operations, agent tool execution, failures, and performance metrics.
- Integrated automated evaluations into CI/CD pipelines to detect regressions across prompt, model, and retrieval changes before deployment.

Image Recognition System
Python, Bash, C, YOLO

- Developed and trained a real-time object detection system using YOLO and convolutional neural networks to identify custom object classes.
- Built Python, Bash, and C scripts to collect and preprocess image datasets for training, testing, and inference.

EDUCATION

Master of Science in Computer Science and Artificial Intelligence (In Progress) **Expected 2028**
Boston University

Bachelor of Science in Computer Science, GPA: 3.4 **2016 - 2019**
CUNY Medgar Evers College, Brooklyn, NY